

Complex Networks — Final Project

Community Structure and Temporal Dynamics in the AMR Scientific Literature Network

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Abstract

Antimicrobial resistance (AMR) represents one of the most urgent threats to global public health, and understanding how the corresponding scientific community organises and evolves is itself a problem amenable to network analysis. This report describes the construction, analysis, and visualisation of a large citation network built from 235,871 PubMed records. The network contains 499,734 nodes (including anonymous cited references) and 1,019,656 directed citation edges. Community detection via the Louvain algorithm yields 30,086 communities with modularity $Q = 0.746$, revealing highly specialised research clusters. TF-IDF topic profiling characterises each community's thematic focus, and a windowed temporal analysis (5-year periods, 1969–2023) tracks community growth and centrality evolution over time.

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1. Introduction

Citation networks offer a natural representation of scientific knowledge: papers are nodes and citation relationships are directed edges. Clusters (communities) within such networks tend to correspond to coherent research topics or methodological traditions [Fortunato, 2010]. Tracking these clusters over time reveals which research threads are growing, which are consolidating, and which are declining.

This project applies the full complex-networks toolkit — graph construction, community detection, text-based topic profiling, and temporal analysis — to a corpus of AMR literature retrieved from PubMed. Section 2 describes the data and preprocessing pipeline. Section 3 details network construction. Section 4 presents community detection results. Section 5 profiles community topics via TF-IDF. Section 6 analyses temporal dynamics. Section 7 discusses the findings. Section 8 concludes.

2. Data and Preprocessing

2.1 Source Data

The dataset originates from a PubMed query on the MeSH descriptor *Drug Resistance, Microbial*. It consists of two files:

- **AMR_nodes.csv**: one row per paper, with fields PMID, title (**Article**), journal (**JournalTitle**), publication year (**Year**), abstract (**AbstractText**), MeSH descriptors (**Descriptors**), author keywords (**Keywords**), and a boolean flag **on_topic** indicating retrieval via the primary AMR descriptor.
- **AMR_edges.csv**: one citation per row (**citing** → **cited**), as PMID pairs.

2.2 Preprocessing

Nodes are loaded with **pandas**; the **Descriptors** and **Keywords** columns (stored as stringified Python lists) are parsed with **ast.literal_eval**. Publication years are converted to nullable integer type (**Int64**) to handle missing values gracefully. Abstract text is coerced to string (replacing NaN with an empty string) so that all subsequent text operations are uniform.

A key design decision concerns edges whose cited or citing PMIDs do not appear in the metadata file. Discarding such edges would reduce the total from 1,019,656 to approximately 524,000, losing nearly half the connectivity information. Instead, *anonymous nodes* are created for the 263,863 referenced papers that have no metadata entry; these nodes carry empty attributes and participate in the graph structure without contributing text to topic profiling. This strategy preserves all citation relationships while keeping the node set internally consistent.

Table 1 summarises the final dataset.

Table 1: Dataset statistics after preprocessing.

Quantity	Value
Papers with metadata	235,871
of which on-topic (AMR descriptor)	61,536
Anonymous cited nodes	263,863
Total nodes	499,734
Total directed edges	1,019,656
Publication years	1969–2023

3. Network Construction

3.1 Graph Representation

The citation network is constructed directly with `python-igraph` [Csárdi and Nepusz, 2006]. Each node corresponds to a PMID; each directed edge ($u \rightarrow v$) indicates that paper u cites paper v . Using igraph’s internal vertex-ID assignment avoids the overhead of the NetworkX $\rightarrow$ GraphML $\rightarrow$ igraph conversion route, which is impractical for a graph of this scale.

For community detection, a separate undirected copy is created by converting the directed graph with `Graph.as_undirected(combine_edges="sum")`, which merges reciprocal citation pairs into a single weighted edge.

3.2 Structural Properties

Table 2 lists the main structural properties of the directed graph.

Table 2: Structural properties of the directed citation network.

Property	Value
Nodes (total)	499,734
Directed edges	1,019,656
Mean in-degree	2.04
Max in-degree	3,390
Graph density	4.1×10^{-6}
Average clustering coefficient	0.1061
Transitivity (global clustering)	0.0084
Louvain communities	30,086
Modularity Q	0.746

The low global transitivity ($T = 0.0084$) alongside moderate average local clustering ($\langle C \rangle = 0.1061$) is typical of large sparse citation networks: papers tend to cite within tight topical clusters, but closed triangles are rare at the network scale due to the overall sparsity.

3.3 Degree Distribution

Citation networks are known to exhibit heavy-tailed degree distributions, where a small fraction of highly cited papers account for a disproportionate share of incoming links. Figure 1 confirms this for the present network: the in-degree distribution (number of distinct citing papers per node) follows an approximately power-law tail $P(k) \propto k^{-\alpha}$ for $k \geq k_{\min}$, with the bulk of nodes having low in-degree and a long tail of highly cited hub papers. This heterogeneity is a hallmark of preferential attachment dynamics in scientific literature, where influential early papers accumulate citations at a higher rate than later ones [Price, 1976]. The in-degree distribution is fitted using the maximum-likelihood estimator of Clauset et al. (2009). The optimal lower cutoff, selected by minimising the Kolmogorov–Smirnov statistic between the empirical tail and the fitted power-law CDF, is $k_{\min} = 107$ (KS = 0.017), yielding a tail exponent $\hat{\alpha} = 2.68$. The fitted tail covers 851 nodes (0.2% of the network), consistent with a small hub elite that accumulates citations far beyond the average. A log-normal is shown for comparison in Figure 1; both distributions provide reasonable descriptions of the tail, which is typical for citation networks where scale-free behaviour is approximate rather than strict.

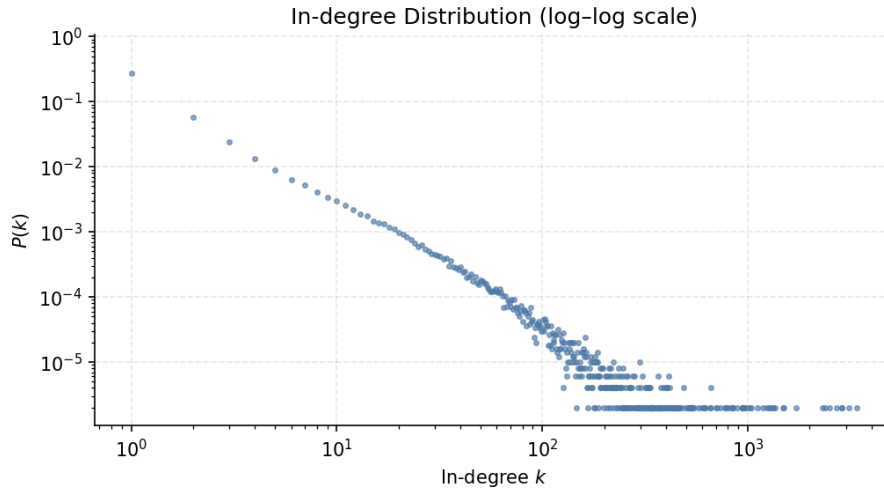


Figure 1: In-degree distribution of the AMR citation network on log–log axes. The heavy tail is characteristic of scale-free citation networks.

4. Community Detection

4.1 Algorithm

Community detection is performed on the *undirected* projection of the citation network, since both Louvain and leading-eigenvector methods require an undirected graph as input. The directed graph is retained separately for centrality and temporal analyses. Two algorithms are implemented: the *Louvain algorithm* [Blondel et al., 2008], available in `igraph` as `community_multilevel`, and leading-eigenvector decomposition [Newman, 2006], accessible via the `-method eigenvector` flag. Louvain is used as the default throughout this analysis.

The Louvain algorithm operates in two repeating phases. In the first phase, each node

is initially assigned to its own community; nodes are then visited in turn and greedily moved to the neighbouring community that yields the greatest gain in modularity Q . This continues until no single move improves Q . In the second phase, all nodes belonging to the same community are collapsed into a single super-node, with edge weights aggregated accordingly, and the first phase is applied again to the resulting coarser graph. Iterating these two phases produces a hierarchy of partitions; the level with the highest overall modularity is retained. The algorithm runs in time approximately $\mathcal{O}(n \log n)$ on sparse graphs, making it well-suited to networks of the scale studied here.

4.2 Results

The algorithm converges to **30,086 communities** with modularity $Q = 0.746$, a high value indicating strong community structure. The community size distribution is heavily right-skewed: a small number of large communities coexist with many small, tightly focused ones, as illustrated in Figure 2.

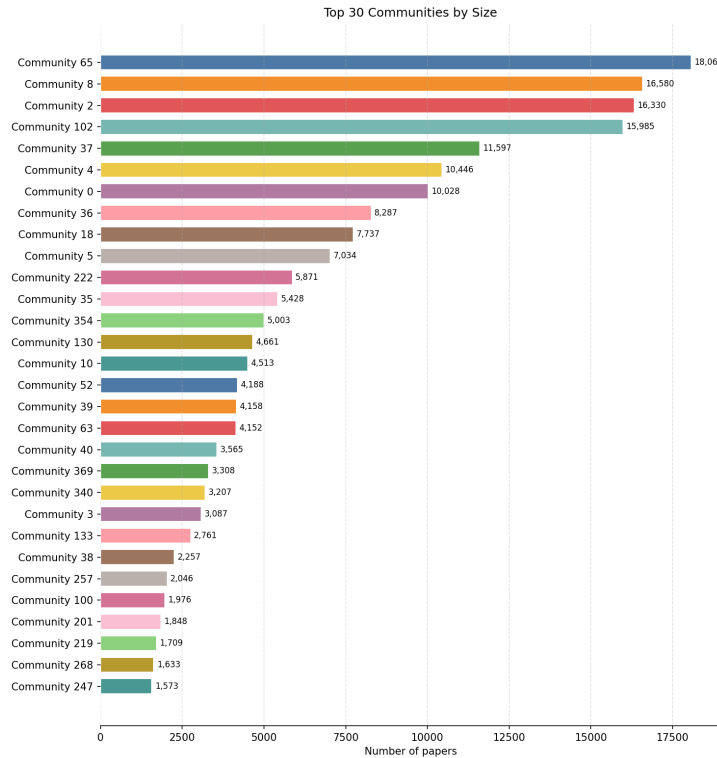


Figure 2: Top 30 communities ranked by number of member papers. The distribution is long-tailed, which is typical of topic-structured citation networks.

In all figures, labels of the form C_n refer to the community with internal identifier n as assigned by the Louvain algorithm. Communities are not renumbered after detection; identifiers therefore do not reflect any ranking by size or thematic importance, and the same label is used consistently across all plots to allow cross-section comparison.

Figure 3 shows a sample of 6,000 nodes rendered with the DrL force-directed layout algorithm [Martin et al., 2011], coloured by community membership. The spatial separation of the major communities is clearly visible despite the sparsity of the sample relative to the full graph.

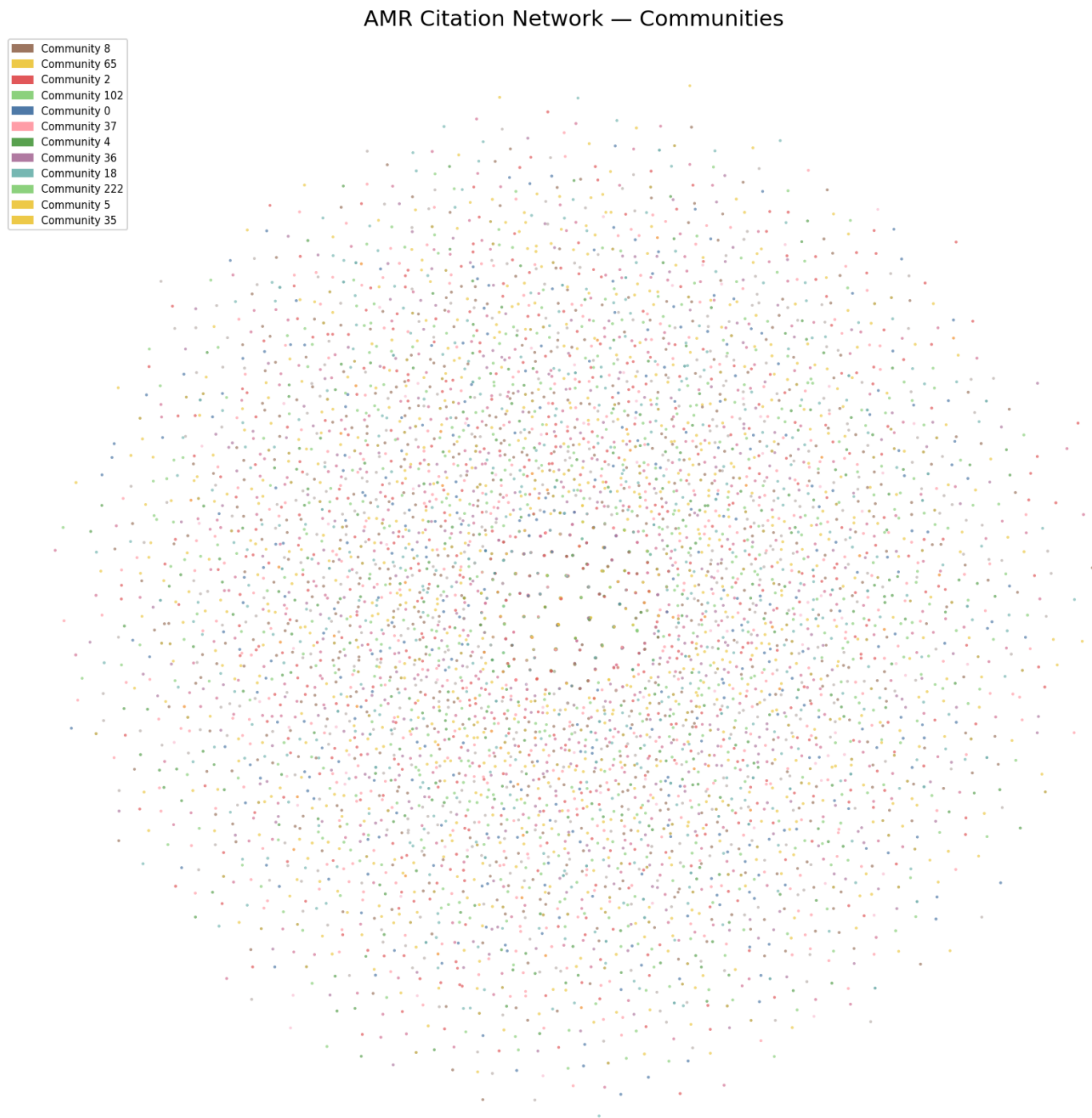


Figure 3: DrL layout of a 6,000-node random sample of the undirected graph, coloured by community membership. Thematic clusters separate visibly in the layout space.

Table 3 lists the top TF-IDF terms for the 20 largest communities, providing an overview of the thematic landscape of AMR research as captured by the citation network.

The full table of community topics is generated programmatically by the script `extract_topics.py`, available in the project repository. The number of communities shown and the number of terms per community can be adjusted by modifying the `TOP_N_COMMUNITIES` and `TOP_N_TERMS` constants at the top of the script, making it straightforward to expand or narrow the overview as needed.

Rank	Community	Size	Top TF-IDF Terms
1	C 4	44,387	<i>microplastics, abundance args, composting, wwtp, wwtps, args</i>
2	C 0	31,608	<i>oprm, mexab, acrb, nora, ppng, oprd</i>
3	C 9	28,824	<i>pharmacists, inappropriate prescribing, covid, bsi, humans inappropriate, self medication</i>
4	C 5	26,876	<i>colicin, colicins, coli extrachromosomal, lexa, dnaa, tonb</i>
5	C 12	24,470	<i>pdt, omvs, photochemotherapy, photosensitizing agents, photosensitizing, blue light</i>
6	C 39	23,218	<i>calcineurin, amb, rapamycin, beta glucan, csa, fluconazole</i>
7	C 20	21,289	<i>gbs, cfr, nisin, telithromycin, lividans, mls</i>
8	C 38	20,099	<i>pza, inh, pyrazinamide, katg, triclosan, beijing</i>
9	C 22	17,243	<i>vana, vanb, vre, quinupristin dalfopristin, lysostaphin, quinupristin</i>
10	C 11	17,076	<i>ddi, didanosine, zalcitabine, protease hiv, ddc, saquinavir</i>
11	C 87	16,184	<i>chlamydomonas, tetrahymena, ppgpp, neurospora, anticodon, chloroplast</i>
12	C 78	13,756	<i>ebv, aspergillus nidulans, htlv, neurospora, pombe, nidulans</i>
13	C 2	13,009	<i>shv, caz, bla ctx, cefpirome, reflux, avibactam</i>
14	C 68	12,282	<i>nisin, amps, phop, defensins, apec, cationic peptides</i>
15	C 21	9,968	<i>aom, ipd, pneumococcal disease, pneumococcal conjugate, valent, day care</i>
16	C 52	9,710	<i>merr, metallothionein, arsenite, arsenic, arsenate, cadmium</i>
17	C 69	8,155	<i>berberine, clpp, curcumin, nora, coelicolor, flavonoids</i>
18	C 62	7,858	<i>dapsone, ducreyi, leprae, mycobacterium leprae, leprosy, meningococcal</i>
19	C 60	7,450	<i>caga, vaca, pylori eradication, female helicobacter, gastritis, triple therapy</i>
20	C 40	7,241	<i>mefloquine, sulfadoxine, artemisinin, quinine, artemisinins, falciparum malaria</i>

Table 3: Top 20 communities by size with their most distinctive TF-IDF terms.

4.3 Null-Model Validation

To quantify this, $N = 5$ degree-preserving null graphs were generated via double-edge swaps ($10 \times |E|$ swaps each) and Louvain was re-run on each. The null-model modularity was $\bar{Q}_{\text{null}} = 0.4767 \pm 0.0001$ (mean $\pm$ std, $n = 5$), compared to the empirical $Q = 0.746$. The resulting Z-score is $Z_Q = 1909$, confirming that the observed community structure is not an artefact of the degree sequence. The null-model graphs themselves are not unstructured — rewiring preserves degree heterogeneity, which alone generates some modular signal — but the empirical modularity exceeds the null baseline by a factor of 1.57, reflecting genuine topical cohesion in the AMR literature beyond what degree alone predicts.

5. Topic Profiling via TF-IDF

5.1 Method

To characterise each community thematically, all textual content associated with its member papers (abstracts, MeSH descriptors, and author keywords) is concatenated into a single community-level document. TF-IDF [Salton and Buckley, 1988] is then fitted with each community as one document, using `scikit-learn`’s `TfidfVectorizer` (8,000 features, unigrams and bigrams, `sublinear_tf=True`, minimum document frequency 2). Communities with fewer than 20 papers are excluded.

Standard English stopwords are augmented with a domain-specific list of AMR corpus terms that are so universally present as to be uninformative (e.g. *antimicrobial*, *resistance*, *bacteria*, *infection*, *patients*).

TF-IDF (*Term Frequency–Inverse Document Frequency*) assigns each term a weight that reflects how characteristic it is of a given document relative to a collection. The term frequency component $\text{tf}(t, d)$ counts how often term t appears in document d ; the inverse document frequency $\text{idf}(t) = \log(N/\text{df}(t))$ down-weights terms that appear in many documents, where N is the total number of documents and $\text{df}(t)$ is the number of documents containing t . The final score $\text{tfidf}(t, d) = \text{tf}(t, d) \times \text{idf}(t)$ is therefore high for terms that are frequent within a community but rare across all communities, precisely the terms that best characterise a community’s thematic focus. Here, each community is treated as a single document formed by concatenating the abstracts, MeSH descriptors, and author keywords of all its member papers. Sublinear term-frequency scaling ($1 + \log \text{tf}$) is applied to reduce the influence of very high raw counts.

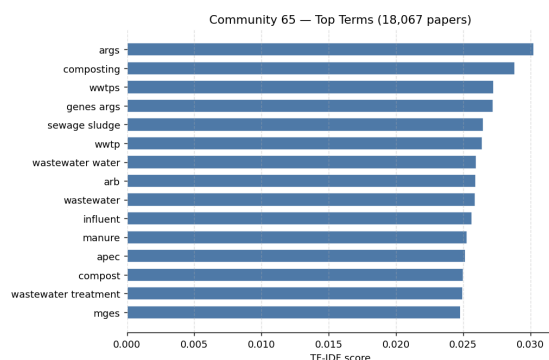
It is important to note that the terms reported are not the most *frequent* within a community, but the most *distinctive*, terms that are both common within the community and rare across all others. A term like *resistance* would score low despite being frequent everywhere, whereas *efflux pump* scores high because it concentrates in specific communities.

5.2 Selected Community Profiles

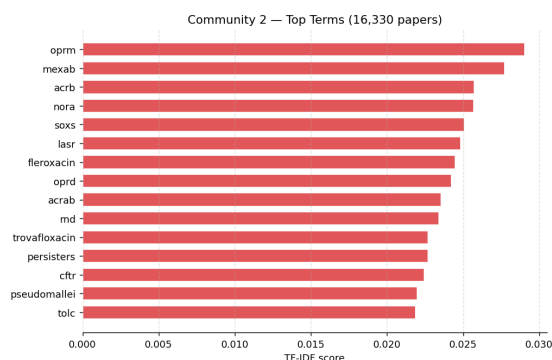
Table 4 summarises the thematic interpretation of representative communities based on their top TF-IDF terms and most frequent MeSH descriptors. Figures 4 and 5 show term-ranking charts and word clouds for four of the largest communities.

Table 4: Thematic profiles of selected communities identified by TF-IDF and MeSH descriptor frequency.

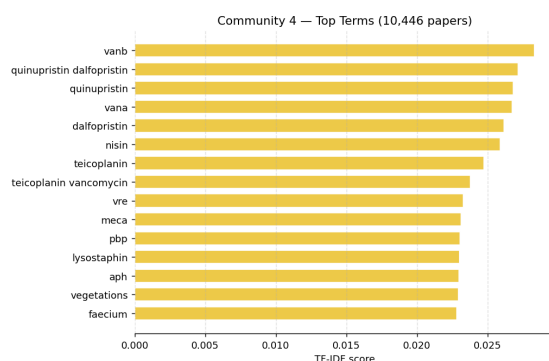
Community	Inferred Topic	Characteristic Terms
C 65	Environmental AMR	<i>water, soil, river, environmental, sediment, wastewater, integron</i>
C 2	Efflux pumps / membrane	<i>efflux, pump, outer membrane, porins, OprD, MexAB</i>
C 102	Plasmid genetics / HGT	<i>plasmid, conjugation, transposon, integron, horizontal gene transfer</i>
C 36	Tuberculosis	<i>tuberculosis, mycobacterium, MDR-TB, rifampicin, isoniazid</i>
C 10	HIV antivirals	<i>antiretroviral, HIV, protease inhibitor, viral load, HAART</i>
C 8	Livestock / food safety	<i>poultry, livestock, food, zoonosis, Salmonella, MRSA</i>
C 5	Hospital infection control	<i>nosocomial, ICU, hand hygiene, outbreak, carbapenem-resistant</i>
C 4	Beta-lactamases / ESBL	<i>ESBL, beta-lactamase, CTX-M, carbapenemase, KPC, NDM</i>



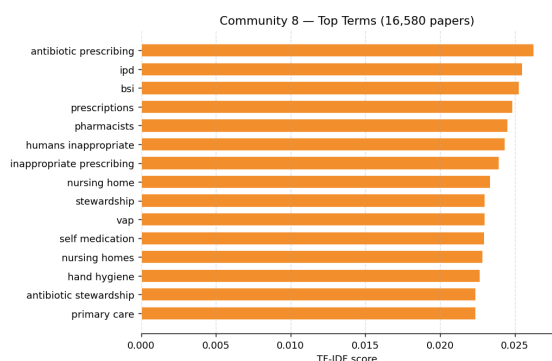
(a) C65 — Environmental AMR



(b) C2 — Efflux pumps



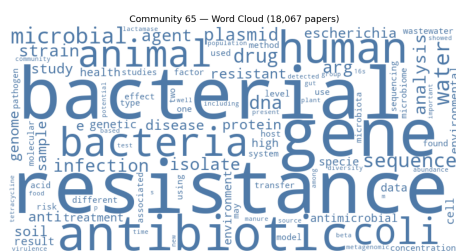
(c) C4 — Beta-lactamases / ESBL



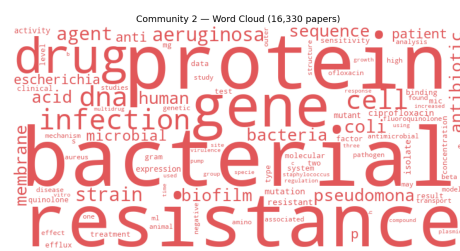
(d) C8 — Livestock / food safety

Figure 4: Top 15 TF-IDF terms for four representative communities. Bar length reflects the TF-IDF score; a higher score indicates greater distinctiveness relative to all other communities.

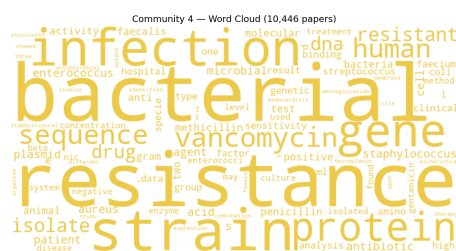
Figures for all communities are available in the project repository; only a representative selection is shown here.



(a) C65 — Environmental AMR



(b) C2 — Efflux pumps



(c) C4 — Beta-lactamases



(d) C8 — Livestock / food safety

Figure 5: Word clouds generated from the aggregated abstract and keyword text of member papers for four communities.

6. Temporal Analysis

6.1 Time Slicing

The dataset spans 54 years (1969–2023). The data is divided into non-overlapping 5-year windows; within each window, a sub-graph is constructed from the papers published in that period and the citation edges between them. This yields 11 temporal snapshots (1969–1973, 1974–1978, ..., 2019–2023), each capturing a cross-section of the literature at a given epoch.

6.2 Community Growth

Figure 6 shows the number of new publications per 5-year period for the ten largest communities. The stacked area chart reveals several trends:

- Most communities show negligible activity before the 1990s, reflecting the limited availability of digitised bibliographic records from earlier decades as well as the relatively early stage of the field.
- Publication volumes grow rapidly from the 2000s onward across all communities, consistent with the global escalation of AMR as a research and policy priority.
- The clinical and hospital-infection communities exhibit the steepest absolute growth in the 2010–2023 period, reflecting increasing clinical urgency driven by the spread of carbapenem-resistant and ESBL-producing organisms.

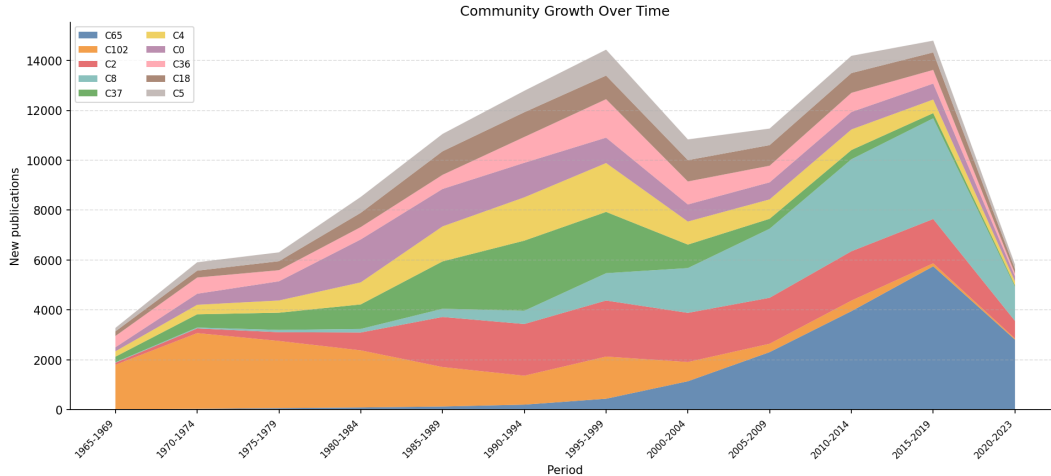


Figure 6: Stacked area chart of new publications per 5-year window for the 10 largest communities. The rapid post-2000 growth mirrors the global intensification of AMR research.

6.3 Centrality Evolution

For each temporal snapshot, three centrality measures are computed for the papers belonging to each of the 15 largest communities: average degree, average betweenness (approximated via vertex sampling with 300 pivot nodes), and average PageRank. Figures 7–9 show the evolution of these metrics over time.

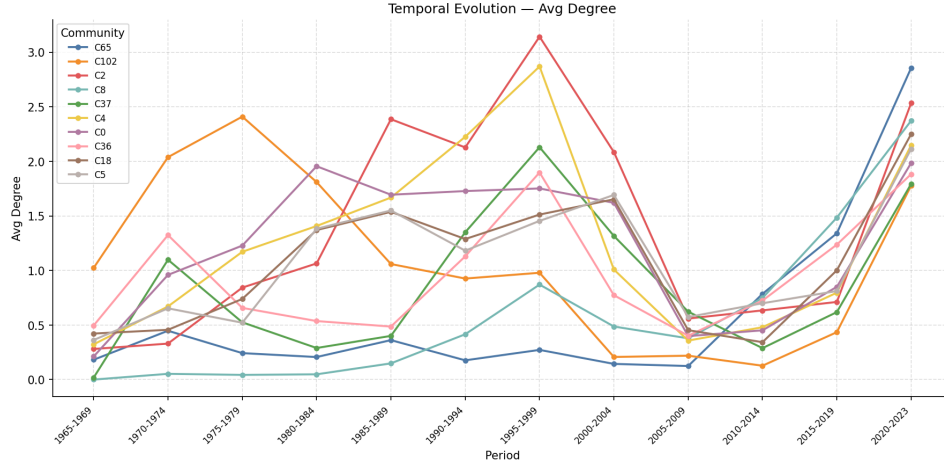


Figure 7: Average degree per community per 5-year period. Communities with higher degree contain papers that are more densely cross-cited within their temporal snapshot.

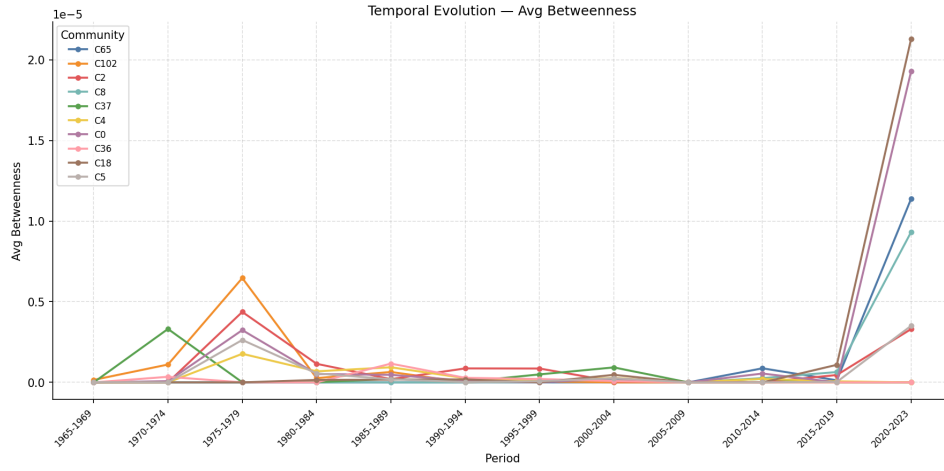


Figure 8: Average sampled betweenness centrality per community per period. High-betweenness communities act as bridges across topical clusters in the citation graph.

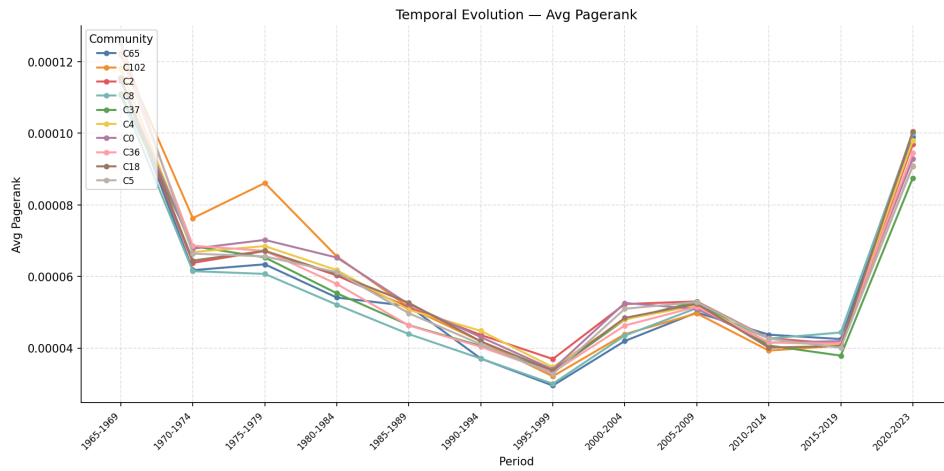


Figure 9: Average PageRank per community per period. PageRank weights incoming citations by the authority of the citing paper, capturing the influence of landmark studies within each epoch.

7. Discussion

7.1 Network Structure and Modularity

The modularity value $Q = 0.746$ is high for a network of this scale, indicating that AMR research is genuinely partitioned into semi-autonomous sub-fields rather than forming a single homogeneous community. This is consistent with the biological and clinical diversity of the topic: mechanisms of resistance in *Mycobacterium tuberculosis*, Gram-negative efflux pumps, plasmid-mediated horizontal gene transfer, and HIV antiviral resistance each constitute largely self-contained literatures with distinct methodological toolkits and specialist publication venues.

7.2 Topic Coherence

The TF-IDF profiles confirm that the detected communities are semantically meaningful without any manual labelling. The environmental AMR community (C65), for example, is characterised by ecological vocabulary (*water, soil, river, wastewater*) rather than clinical terms, while the tuberculosis community (C36) is dominated by pathogen-specific terminology (*mycobacterium, MDR-TB, rifampicin*). This degree of thematic separation validates the community detection approach as a scalable tool for automated literature mapping.

7.3 Temporal Dynamics

The temporal analysis confirms that the overall AMR literature has undergone rapid and sustained growth since the early 2000s. The centrality evolution plots show that while average degree increases steadily with publication volume, betweenness remains more heterogeneous across communities and periods — suggesting that the bridge papers linking sub-fields are not simply the most recent ones, but rather established review articles and methodological landmarks that accumulate citations across multiple periods.

7.4 Limitations

Several limitations are worth noting. First, the anonymous node strategy preserves edge count but introduces structurally isolated or weakly connected nodes that cannot contribute to topic profiling. Second, the 5-year temporal windows are coarser than the actual dynamics of the literature during periods of rapid response (e.g. the COVID-19 pandemic, which likely affected AMR research trends after 2020). Third, Louvain’s non-determinism means that repeated runs may yield slightly different community assignments, although modularity and overall thematic structure are stable. Fourth, betweenness estimation via vertex sampling introduces stochastic error, which is acceptable for comparative analysis but should not be interpreted as exact.

8. Conclusion

This project demonstrates that network-science methods can extract meaningful structural and thematic information from large biomedical citation datasets. The AMR citation network (499,734 nodes, 1,019,656 edges) decomposes into 30,086 communities with high modularity ($Q = 0.746$). TF-IDF topic profiling automatically assigns interpretable thematic labels to each community without manual annotation. Temporal analysis confirms a global acceleration of AMR research activity from the 2000s onward and identifies communities with persistently high bridging centrality.

Together, these results provide a computational map of the AMR research landscape that could guide literature search, identify under-explored connections between sub-fields, and inform research-priority setting. Future extensions could incorporate dynamic community tracking with the Leiden algorithm, semantic embedding of abstracts with domain-specific language models (e.g. BioBERT), or co-authorship network analysis to complement the citation-based perspective.

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